11.31 Given E_ρ and B_z, $\vec{D}$ and $\vec{H}$ in the cylinder are given by

$$\vec{D} = \varepsilon\vec{E} + \gamma^2\left(\varepsilon - \frac{1}{\mu}\right)\left[\beta^2\vec{E}_\perp + \vec{\beta}\times\vec{B}\right] = \hat{\rho}\left[\gamma^2\left(\varepsilon - \frac{\beta^2}{\mu}\right)E_\rho + \beta\gamma^2\left(\varepsilon - \frac{1}{\mu}\right)B_z\right],$$

$$\vec{H} = \frac{1}{\mu}\vec{B} + \gamma^2\left(\varepsilon - \frac{1}{\mu}\right)\left[-\beta^2\vec{B}_\perp + \vec{\beta}\times\vec{E}\right] = \hat{z}\left[-\gamma^2\left(\beta^2\varepsilon - \frac{1}{\mu}\right)B_z - \beta\gamma^2\left(\varepsilon - \frac{1}{\mu}\right)E_\rho\right].$$

To determine unknown E_ρ and B_z, we need to apply the boundary condition implied from $\nabla\cdot\vec{D} = 0$ and $\nabla\times\vec{H} = 0$, in the reference frame where the cylinder is at rest, i.e. K'. Since $(\vec{D}, \vec{H})$ transforms as a pair in the same way as $(\vec{E}, \vec{B})$, in K' outside of the cylinder,

$$\vec{D}'_{out} = \gamma\left(\vec{D}_{out} + \vec{\beta}\times\vec{H}_{out}\right) - \frac{\gamma^2}{\gamma+1}\vec{\beta}(\vec{\beta}\cdot\vec{D}_{out}) = \gamma\beta B_0\hat{\rho}.$$

$$\vec{H}'_{out} = \gamma\left(\vec{H}_{out} - \vec{\beta}\times\vec{D}_{out}\right) - \frac{\gamma^2}{\gamma+1}\vec{\beta}(\vec{\beta}\cdot\vec{H}_{out}) = \gamma B_0\hat{z}.$$

NOT NECESSARY!

Since $\vec{D}_{out} = 0$, $\vec{H}_{out} = B_0\hat{z}$. Inside the cylinder, we have similarly

$$\vec{D}'_{in} = \gamma\left(\vec{D}_{in} + \vec{\beta}\times\vec{H}_{in}\right) - \frac{\gamma^2}{\gamma+1}\vec{\beta}(\vec{\beta}\cdot\vec{D}_{in}) = \gamma\left(D_\rho + \beta H_z\right)\hat{\rho}$$

$$\vec{H}'_{in} = \gamma\left(\vec{H}_{in} - \vec{\beta}\times\vec{D}_{in}\right) - \frac{\gamma^2}{\gamma+1}\vec{\beta}(\vec{\beta}\cdot\vec{H}_{in}) = \gamma\left(H_z + \beta D_\rho\right)\hat{z},$$

since $\vec{\beta}$ is normal to both $\vec{D}$ and $\vec{H}$. The continuity in the normal component of $\vec{D}$ and tangent component of $\vec{H}$ lead to

$$\gamma\left(D_\rho + \beta H_z\right) = \gamma\beta B_0, \quad \gamma\left(H_z + \beta D_\rho\right) = \gamma B_0$$

which is

$$D_\rho = 0, \quad H_z = B_0.$$

From the first condition, we have

$$E_\rho = -\frac{\beta(\varepsilon - 1/\mu)B_z}{\varepsilon - \beta^2/\mu}$$

Substitute it into the second condition, we will obtain

$$\left[-\left(\beta^2\varepsilon - \frac{1}{\mu}\right) + \beta\left(\varepsilon - \frac{1}{\mu}\right)\frac{\beta(\varepsilon - 1/\mu)}{\varepsilon - \beta^2/\mu}\right]\gamma^2 B_z = B_0.$$

or $$\frac{\varepsilon - \beta^2/\mu}{\gamma^2}B_0 = \left[\beta^2\left(\varepsilon - \frac{1}{\mu}\right)^2 - \left(\beta^2\varepsilon - \frac{1}{\mu}\right)\left(\varepsilon - \beta^2/\mu\right)\right]B_z = \frac{\varepsilon}{\mu}(1-\beta^2)^2 B_z = \frac{\varepsilon}{\mu}(1-\beta^2)\frac{B_z}{\gamma^2}.$$

Then, $B_z = \mu B_0 \dfrac{1-\beta^2/\mu\varepsilon}{1-\beta^2} = \mu B_0 \dfrac{1-\omega^2\rho^2/c^2\mu\varepsilon}{1-\omega^2\rho^2/c^2}$,

and $E_\rho = -\dfrac{\beta(1-1/\mu\varepsilon)}{1-\beta^2/\mu\varepsilon} B_z = -\dfrac{\beta(1-1/\mu\varepsilon)}{1-\beta^2}\mu B_0 = -\dfrac{\mu\omega\rho B_0}{c(1-\omega^2\rho^2/c^2)}\left(1-\dfrac{1}{\varepsilon\mu}\right)$.

The voltage difference now is

$$V = \int_a^b E_\rho \, d\rho = -\frac{\mu\omega B_0}{c}\left(1-\frac{1}{\mu\varepsilon}\right)\int_a^b \frac{\rho\, d\rho}{1-\omega^2\rho^2/c^2}$$

$$= -\frac{\mu\omega B_0}{c}\left(1-\frac{1}{\mu\varepsilon}\right)\cdot\left(-\frac{1}{2}\frac{c^2}{\omega^2}\log\left(1-\omega^2\rho^2/c^2\right)\right)\Big|_a^b$$

$$= \frac{\mu c B_0}{2\omega}\left(1-\frac{1}{\mu\varepsilon}\right)\log\left(\frac{1-\omega^2 b^2/c^2}{1-\omega^2 a^2/c^2}\right)$$

For non-relativistic limit, $\log(1-\omega^2\rho^2/c^2) \approx -\omega^2\rho^2/c^2$, and

$$|V| = \frac{\mu\omega}{2c} B_0\left(1-\frac{1}{\mu\varepsilon}\right)(b^2-a^2).$$